Statistical Modeling Support Group
PART A: THEORY MCQ PRACTICE PAPER

1. A lecturer records test marks for the same 18 students before and after a revision workshop. Students are independent of one another, and the within-student differences are approximately normal. Which method directly tests whether the population mean change is zero?

(A) The pooled t-test applied to two independent groups of students.

(B) The paired t-test applied to each student's before-and-after score difference.

(C) Binary logistic regression applied to whether each student passed training.

(D) One-way ANOVA applied to three independent groups of different students.

(E) Pearson correlation analysis applied to associations between two numerical predictors.

2. A manufacturer compares mean battery lifetimes from two independent normal populations. There are 12 batteries of one brand and 17 of another; population variances are unknown and markedly unequal. Which method is best suited to testing a difference between the population means?

(A) Use pooled testing with common variance and combined reference degrees.

(B) Use Welch testing with separate variances and approximate reference degrees.

(C) Use paired testing with matched measurements and difference-based reference degrees.

(D) Use normal testing with known variances and fixed population deviations.

(E) Use variance-ratio testing with sample variances and ordered reference degrees.

3. Two independent groups receive different teaching methods. Test marks are normal and a common unknown population variance is justified. With sample sizes 14 and 18, which procedure uses the stated common-variance estimator to test the difference in mean marks?

(A) Use a paired two-sample t-test with 13 reference degrees.

(B) Use a variance-ratio two-sample F-test with ordered sample degrees.

(C) Use a pooled two-sample t-test with 32 reference degrees.

(D) Use a pooled two-sample t-test with 30 reference degrees.

(E) Use a known-variance two-sample z-test with specified population deviations.

4. A two-sided 95% confidence interval for $\mu_A - \mu_B$ is $(-1.2, 4.8)$. Using the corresponding test at the 5% level, what follows?

(A) The evidence establishes that both population means are exactly equal.

(B) The evidence establishes a negative difference between both population means.

(C) The interval assigns ninety-five percent probability to exact population equality.

(D) The evidence does not establish a difference between population means.

(E) The evidence establishes a positive difference between both population means.

5. At two independent campuses, a random sample of 400 students gives 288 passes and a sample of 500 gives 390 passes. The task is to test equality of the population pass probabilities, without adjusting for other predictors. Counts are large enough for a normal approximation. Which method directly addresses this question?

(A) A one-way ANOVA comparing average marks across independent teaching groups.

(B) A paired t-test comparing mean scores within matched student pairs.

(C) A variance-ratio F-test comparing mark variability between independent student groups.

(D) A Pearson correlation test comparing associations between numerical study measures.

(E) A two-proportion z-test comparing pass probabilities between independent student groups.

6. A filling machine has an unknown mean output. An engineer samples 16 independent, normally distributed fill amounts and wants to test whether their population variance exceeds 4 square millilitres. Which analysis directly tests the stated claim?

(A) A one-sample Student t-test of the normal population mean.

(B) A two-sample variance-ratio F-test of two normal population variances.

(C) A one-sample proportion z-test of the population defect probability.

(D) A paired Student t-test of the mean measurement difference.

(E) A one-sample chi-square test of the normal population variance.

7. A factory compares the consistency of two independent bottling machines. Fill amounts are normally distributed; their population means are unknown and are not the comparison of interest. Which method directly tests equality of their population variances?

(A) A paired two-sample t-test using the within-bottle measurement differences.

(B) A variance-ratio F-test using both independently estimated sample variances.

(C) A two-proportion z-test using the observed bottle defect counts.

(D) A Pearson correlation test using the paired output values.

(E) A pooled two-sample t-test using a common estimated variance.

8. A crop scientist randomly assigns 24 comparable plots to four fertilizers, six plots per fertilizer. Yield is continuous, and independent normal errors with equal variances are plausible. Which omnibus method tests whether all four population mean yields are equal?

(A) Variance-ratio testing comparing only two independently estimated treatment variances.

(B) One-way ANOVA comparing treatment mean variation against within-treatment variation.

(C) Pearson correlation testing comparing associations between two predictor variables.

(D) Binary logistic regression comparing probabilities of a binary outcome.

(E) Paired t-testing comparing matched yield differences between repeated measurements.

9. After a significant ANOVA, Fisher's LSD is used for many pairwise comparisons. Which caution is correct?

(A) Individual comparison error control need not ensure overall familywise control.

(B) Individual comparison error control guarantees equality for every nonsignificant pair.

(C) Individual comparison error control requires identical sample means across treatments.

(D) Individual comparison error control always ensures exact overall familywise control.

(E) Individual comparison error control removes all uncertainty about treatment rankings.

10. A large randomized study finds a reduction of 0.2 marks on a 100-mark scale with $p < 0.001$. Which assessment is most defensible?

(A) Statistical significance alone cannot establish that this effect matters practically.

(B) Statistical significance alone establishes that this effect matters practically everywhere.

(C) Statistical significance alone establishes that this effect benefits every student.

(D) Statistical significance alone establishes that this effect has no uncertainty.

(E) Statistical significance alone measures the improvement obtained by every student.

11. A regression contains an intercept and a four-level nominal predictor. How should the nominal predictor be encoded to avoid exact collinearity?

(A) Include two indicator variables and merge two categories without justification.

(B) Include one numerical variable and treat category labels as distances.

(C) Include five indicator variables and leave two categories as reference.

(D) Include four indicator variables and leave no category as reference.

(E) Include three indicator variables and leave one category as reference.

12. To compute the VIF for predictor X_j, which auxiliary regression supplies the required R_j^2?

(A) Regress the response on only that single available predictor variable.

(B) Regress fitted values on all the other available predictor variables.

(C) Regress that predictor on all the other available predictor variables.

(D) Regress squared residuals on only that single available predictor variable.

(E) Regress the response on all predictors after removing that predictor.

13. An irrelevant predictor is added to an ordinary least-squares model fitted to the same observations, with an intercept. What can happen?

(A) Training R^2 cannot decrease while adjusted R^2 must increase.

(B) Training R^2 cannot decrease while adjusted R^2 can decrease.

(C) Training R^2 must decrease while adjusted R^2 must decrease.

(D) Training R^2 remains fixed while adjusted R^2 remains fixed.

(E) Training R^2 must decrease while adjusted R^2 must increase.

14. A residual-versus-fitted plot has a clear funnel shape, with spread increasing as fitted values increase. Which assumption is most directly questioned?

(A) The predictor variables have identical means across all fitted values.

(B) The conditional error mean remains constant across all fitted values.

(C) The regression error distribution remains normal across all fitted values.

(D) The conditional error variance remains constant across all fitted values.

(E) The response variable has identical means across all fitted values.

15. A business fits a linear regression to measurements recorded in a meaningful time order. It wants a diagnostic test specifically for first-order serial correlation in the regression errors. Which method should it select?

(A) The Shapiro–Wilk test for normality of the observed regression residuals.

(B) The Hosmer–Lemeshow test for grouped calibration of binary outcome probabilities.

(C) The Durbin–Watson test for first-order correlation in ordered regression errors.

(D) The Levene test for equality of variances across independent groups.

(E) The variance inflation factor for linear dependence among predictor variables.

16. An analyst selects the best model from 100 candidates using a test set, then reports the same test-set error. What is the main problem?

(A) The reported error becomes zero because selection reused evaluation observations.

(B) The reported error estimates causality because testing excluded training observations.

(C) The reported error may be optimistic because selection reused evaluation.

(D) The reported error must be unbiased because testing excluded training.

(E) The reported error must increase because selection reused evaluation observations.

17. With standardized predictors and a positive tuning penalty, which distinction between ridge and lasso is generally correct?

(A) Lasso penalizes squared coefficients while ridge penalizes absolute coefficient values.

(B) Ridge always removes correlation while lasso always preserves coefficient magnitudes.

(C) Ridge can produce exact zeros while lasso generally retains predictors.

(D) Lasso and ridge always retain exactly identical predictor variable subsets.

(E) Lasso can produce exact zeros while ridge generally retains predictors.

18. For a full-column-rank design matrix X, which expression gives the ordinary least-squares coefficient estimator?

(A) The coefficient vector is obtained as $X(X^TX)^{-1}y$.

(B) The coefficient vector is obtained as $(X^TX)X^Ty$.

(C) The coefficient vector is obtained as $(X^TX)^{-1}Xy$.

(D) The coefficient vector is obtained as $(X^TX)^{-1}X^Ty$.

(E) The coefficient vector is obtained as $(XX^T)^{-1}X^Ty$.

19. A health researcher wants to model systolic and diastolic blood pressure jointly using age and body mass index. Both responses are continuous. Which model description matches this task?

(A) Multivariate multiple regression with two responses and two predictor variables.

(B) Ordinary multiple regression with one response and two predictor variables.

(C) Multinomial logistic regression with one response and two predictor variables.

(D) Binary logistic regression with one response and two predictor variables.

(E) Simple linear regression with one response and one predictor variable.

20. A scatterplot suggests that mean fuel consumption curves smoothly with vehicle speed. The response is continuous, and a quadratic mean with independent normal, constant-variance errors is plausible. Which candidate directly represents this curved mean while remaining linear in its unknown coefficients?

(A) Ordinal logistic regression using X and X^2 as predictor terms.

(B) Simple linear regression using only X as the predictor term.

(C) Polynomial linear regression using X and X^2 as predictor terms.

(D) Poisson count regression using X and X^2 as predictor terms.

(E) Binary logistic regression using X and X^2 as predictor terms.

21. In a generalized linear model, what is connected to the linear predictor through the link function?

(A) The individual observed response after removing all fitted predictor effects.

(B) The conditional predictor variance given the observed values of responses.

(C) The marginal predictor mean given the observed values of responses.

(D) The unconditional residual variance after removing all fitted predictor effects.

(E) The conditional response mean given the observed values of predictors.

22. A university wants to estimate whether a student will fail (1) or pass (0), adjusted for attendance, prior GPA and study hours. It wants a model directly linear in the event log-odds without assuming normally distributed predictors within each class. Which method best matches the objective?

(A) Binary logistic regression modelling failure log-odds with all specified predictors.

(B) Poisson count regression modelling failure counts with all specified predictors.

(C) Gamma response regression modelling failure costs with all specified predictors.

(D) Ordinary linear regression modelling failure scores with all specified predictors.

(E) Linear discriminant analysis assuming Gaussian predictors within both outcome classes.

23. A fitted binary logistic model gives linear predictor $\eta = \log(3)$. What is the predicted event probability?

(A) The fitted event probability equals 0.75 for this observation.

(B) The fitted event probability equals 0.50 for this observation.

(C) The fitted event probability equals 0.33 for this observation.

(D) The fitted event probability equals 3.00 for this observation.

(E) The fitted event probability equals 0.25 for this observation.

24. A predictor has coefficient $\log(1.5)$ in a binary logistic model. Holding all other predictors constant, what does a one-unit increase imply?

(A) The event odds increase by one hundred percent from baseline.

(B) The event probability increases by fifty percent from its baseline.

(C) The event odds increase by fifty percent from their baseline.

(D) The event probability increases by fifty percentage points from baseline.

(E) The event odds decrease by fifty percent from their baseline.

25. A Wald 95% confidence interval for a logistic coefficient is $(-0.2, 0.6)$. Which conclusion about its odds-ratio interval follows?

(A) The corresponding odds-ratio interval excludes zero and remains entirely negative.

(B) The corresponding odds-ratio interval excludes one and remains entirely positive.

(C) The corresponding odds-ratio interval includes one and remains entirely positive.

(D) The corresponding odds-ratio interval includes zero and remains partly negative.

(E) The corresponding odds-ratio interval includes one and also includes zero.

26. Two nested logistic models use the same observations. The reduced deviance is 120 and the full deviance is 112 after adding two coefficients. Which LRT reference applies?

(A) Compare the deviance reduction 8 against chi-square with 2 degrees.

(B) Compare the deviance sum 232 against chi-square with 1 degree.

(C) Compare the full deviance 112 against chi-square with 2 degrees.

(D) Compare the deviance reduction 8 against normal with 2 degrees.

(E) Compare the deviance reduction 8 against chi-square with 1 degree.

27. A hospital has fitted a binary logistic model. It wants a grouped goodness-of-fit test that compares observed event counts with counts expected from grouped predicted probabilities. Which listed diagnostic addresses that specific objective?

(A) The Shapiro–Wilk test comparing regression residuals with a normal distribution.

(B) The Hosmer–Lemeshow test comparing grouped observed and expected event counts.

(C) The variance inflation factor comparing predictor dependence through auxiliary regressions.

(D) The Durbin–Watson test comparing adjacent values of ordered regression residuals.

(E) The Levene test comparing response variances across independent treatment groups.

28. Lowering a fixed binary classification threshold from 0.5 to 0.3 on the same predicted probabilities has which consequence?

(A) The number classified negative cannot decrease under the lower threshold.

(B) The number classified positive must halve under the lower threshold.

(C) The number classified positive cannot increase under the lower threshold.

(D) The number classified positive must double under the lower threshold.

(E) The number classified positive cannot decrease under the lower threshold.

29. What is the defining conditional-independence assumption of a Naive Bayes classifier?

(A) Class labels are mutually independent conditional on each predictor value.

(B) Predictor features are mutually dependent conditional on every class label.

(C) Predictor features have identical distributions conditional on every class label.

(D) Predictor features are mutually independent conditional on the class label.

(E) Predictor features are mutually independent without conditioning on any class.

30. A categorical predictor has three possible levels. A class contains five observations, none at level L. With add-one Laplace smoothing, what is $P(L \mid C)$?

(A) The estimated conditional probability becomes 1/3 after add-one smoothing.

(B) The estimated conditional probability becomes 1/6 after add-one smoothing.

(C) The estimated conditional probability becomes 1/5 after add-one smoothing.

(D) The estimated conditional probability becomes 1/8 after add-one smoothing.

(E) The estimated conditional probability becomes 0/5 after add-one smoothing.

31. Factories A and B supply 60% and 40% of components. Their defect rates are 2% and 5%, respectively. Given a defective component, what is the probability it came from B?

(A) The required posterior probability is 0.020 for factory B.

(B) The required posterior probability is 0.050 for factory B.

(C) The required posterior probability is 0.400 for factory B.

(D) The required posterior probability is 0.375 for factory B.

(E) The required posterior probability is 0.625 for factory B.

32. A botanist classifies flowers into three species from four measurements. Within-class multivariate normality is plausible, and the classes are assumed to share one covariance matrix with nonzero correlations between predictors. The model should use class densities and prior probabilities to obtain posterior class probabilities. Which method fits these assumptions?

(A) Principal component analysis constructing directions that maximize overall predictor variance.

(B) Quadratic discriminant analysis using separate covariance estimates within each class.

(C) Multinomial logistic regression directly modelling ratios between outcome category probabilities.

(D) Linear discriminant analysis using one pooled covariance across all classes.

(E) Gaussian Naive Bayes using conditionally independent features within each class.

33. A large labelled sensor dataset has approximately multivariate normal features within each fault class, but class covariance matrices differ substantially, with nonzero feature correlations within classes. There are enough observations to estimate each covariance matrix reliably. Which Gaussian discriminant method explicitly accommodates this difference?

(A) Gaussian Naive Bayes assuming conditional independence among all predictor features.

(B) Principal component analysis ignoring fault labels when constructing variance directions.

(C) Ordinary linear regression treating the fault labels as continuous measurements.

(D) Quadratic discriminant analysis using a separate covariance within each class.

(E) Linear discriminant analysis using a common covariance across all classes.

34. A commuter chooses exactly one of car, bus, train or bicycle. A researcher wants to model all four outcome probabilities jointly using income and travel distance. There is no natural ordering of the transport categories. Which regression method best preserves this response structure?

(A) Poisson count regression treating transport labels as observed event counts.

(B) Ordinal logistic regression using three cumulative logits with shared slopes.

(C) Multinomial logistic regression using three logits against one reference category.

(D) Ordinary linear regression assigning arbitrary numerical scores to transport categories.

(E) Binary logistic regression collapsing all transport modes into two categories.

35. With car as the reference outcome, the income coefficient for bus versus car is positive. Holding other predictors constant, what follows?

(A) The bus probability necessarily increases when the income predictor increases.

(B) The bus-to-car probability ratio increases when the income predictor increases.

(C) The bus-to-train probability ratio increases when the income predictor increases.

(D) The car probability necessarily decreases when the income predictor increases.

(E) The bus-to-car probability ratio decreases when the income predictor increases.

36. A survey records satisfaction as Low, Medium or High. The categories have a meaningful order but no known equal spacing. The researcher assumes the same predictor effect at both cumulative cutpoints. Which regression method directly uses these features?

(A) Proportional-odds ordinal logistic regression using cumulative logits with shared slopes.

(B) Ordinary linear regression using equally spaced scores for satisfaction categories.

(C) Binary logistic regression merging two satisfaction categories into one outcome.

(D) Baseline-category multinomial logistic regression using separate logits with separate slopes.

(E) Poisson count regression interpreting ordered satisfaction scores as event counts.

37. A retailer models the number of complaints in each equal-length observation period using staffing and customer volume. Counts include zero. Conditional on the predictors, a Poisson response distribution is plausible. Which model naturally respects the stated response and distribution?

(A) A binary logistic regression model using a logit mean link.

(B) An ordinal logistic regression model using shared cumulative logit slopes.

(C) A Poisson generalized linear model using a log mean link.

(D) A Gamma generalized linear model using a log mean link.

(E) A normal generalized linear model using an identity mean link.

38. Customers arrive independently at a constant rate of four per hour under a homogeneous Poisson process. A manager wants a probability model for the continuous time until the next customer arrives. Which model and mean are implied by these assumptions?

(A) A Poisson count model with expected wait of 15 minutes.

(B) A Bernoulli event model with expected wait of 4 minutes.

(C) An Exponential waiting-time model with expected wait of 15 minutes.

(D) A normal waiting-time model with expected wait of 15 minutes.

(E) An Exponential waiting-time model with expected wait of 4 minutes.

39. An insurer models strictly positive, continuous claim costs using customer characteristics. Conditional costs are right-skewed, and their variance is approximately $0.4\mu^2$ when the conditional mean is μ. Which candidate directly matches this mean-variance specification?

(A) A Gamma regression with log link and dispersion parameter 0.4.

(B) An Exponential regression with log link and unit squared-mean dispersion.

(C) A normal regression with identity link and constant error variance.

(D) A Poisson regression with log link and unit dispersion parameter.

(E) A binary logistic regression with logit link and Bernoulli variance.

40. A Poisson model with log link estimates a treatment coefficient of $\log(0.8)$. Exposure is equal. What is the adjusted treatment effect?

(A) The expected count decreases by eighty percent relative to control.

(B) The expected count remains completely unchanged when compared with control.

(C) The expected count increases by twenty percent relative to control.

(D) The expected count decreases by twenty percent relative to control.

(E) The expected count increases by eighty percent relative to control.

41. A researcher models a binary outcome and wants to combine a stated normal prior on each logistic coefficient with the current data likelihood, then quantify posterior probabilities about coefficient signs. Which analysis directly implements that objective?

(A) Ordinary linear regression minimizing squared residuals for numerical response values.

(B) Naive Bayes classification multiplying conditional feature likelihoods by class frequencies.

(C) Linear discriminant analysis estimating class densities with a common covariance.

(D) Ordinary logistic regression maximizing likelihood without using coefficient prior distributions.

(E) Bayesian logistic regression combining prior coefficient distributions with Bernoulli likelihood.

42. A Bayesian analysis reports $P(\beta_1 > 0 \mid D) = 0.98$. Which interpretation is valid under its model and prior?

(A) The posterior assigns probability 0.98 to a positive coefficient value.

(B) The repeated-sampling confidence procedure covers positive coefficients in 98% samples.

(C) The posterior assigns probability 0.98 to an exactly zero coefficient.

(D) The predictor increases every individual probability by exactly 0.98 units.

(E) The null hypothesis has sampling probability 0.02 after observing data.

43. Four classrooms are randomized, two per teaching method; 30 students are measured in each classroom. What is the experimental unit?

(A) The teaching method is the unit independently assigned classroom measurements.

(B) The recorded mark is the unit independently assigned teaching treatment.

(C) The examination is the unit independently assigned a teaching treatment.

(D) The classroom is the unit independently assigned a teaching treatment.

(E) The student is the unit independently assigned a teaching treatment.

44. An agricultural field has four soil zones that differ in fertility. Each zone contains three comparable plots, and all three fertilizers must be compared within every zone. Which allocation best uses the known nuisance factor while preserving treatment randomization?

(A) Assign fertilizer according to soil fertility without randomizing individual plots.

(B) Randomize all three fertilizers independently within each complete soil block.

(C) Apply every fertilizer sequentially within each plot without randomizing order.

(D) Apply identical fertilizer positions within every block without randomizing positions.

(E) Assign each fertilizer exclusively to one entire preselected soil block.

45. A monitoring system has 40 correlated continuous sensor measurements with different units. Before visualization, an engineer wants a few uncorrelated linear combinations retaining substantial predictor variance, with no response labels involved. Which method most directly addresses that objective?

(A) Apply LDA to centered measurements grouped by known response categories.

(B) Apply multinomial regression to measurements using observed categorical response values.

(C) Apply PCA to centered measurements scaled by their standard deviations.

(D) Apply Pearson testing separately to every possible sensor measurement pair.

(E) Apply lasso regression to measurements using observed numerical response values.

46. The covariance eigenvalues are $5, 2, 1$. What proportion of total variance is explained by the first two principal components together?

(A) The cumulative explained variance is 25.0% using two components.

(B) The cumulative explained variance is 62.5% using two components.

(C) The cumulative explained variance is 75.0% using two components.

(D) The cumulative explained variance is 12.5% using two components.

(E) The cumulative explained variance is 87.5% using two components.

47. For ordinary PCA of centered data, which relationship between distinct principal-component scores is guaranteed by the construction?

(A) Different component scores have identical loadings but are not independent.

(B) Different component scores are uncorrelated but are not necessarily independent.

(C) Different component scores are correlated but are not necessarily independent.

(D) Different component scores are independent but are not necessarily uncorrelated.

(E) Different component scores have equal variances but are not uncorrelated.

48. A retailer records positive monthly sales. Seasonal fluctuations become proportionally larger as the sales level grows. Which classical decomposition strategy best reflects that pattern?

(A) Use additive decomposition because seasonal amplitude remains constant across levels.

(B) Use multiplicative decomposition because seasonal amplitude scales with series level.

(C) Use a stationary white-noise model because calendar seasonality is absent.

(D) Use ordinary linear regression because all monthly observations are independent.

(E) Use a constant-mean moving-average model because seasonal effects cannot change.

49. A monthly series is differenced once ordinarily and once seasonally at lag 12. Which expression is the resulting series?

(A) The transformed observation is $y_t + y_{t-1} - y_{t-12} + y_{t-13}$.

(B) The transformed observation is $y_t - y_{t-1} - y_{t-12} - y_{t-13}$.

(C) The transformed observation is $y_t - y_{t-1} + y_{t-12} - y_{t-13}$.

(D) The transformed observation is $y_t - 2y_{t-12} + y_{t-24}$.

(E) The transformed observation is $y_t - y_{t-1} - y_{t-12} + y_{t-13}$.

50. After exactly one lag-12 seasonal difference, a monthly series is adequately represented by a nonseasonal ARMA(1,1) model, with no additional ordinary differencing or seasonal AR/MA terms. Which model describes the original, undifferenced series?

(A) Fit ARIMA$(1,0,1)(0,1,0)_{12}$ directly to the original monthly response series.

(B) Fit ARIMA$(1,0,1)(0,2,0)_{12}$ directly to the original monthly response series.

(C) Fit ARIMA$(0,0,1)(1,1,0)_{12}$ directly to the original monthly response series.

(D) Fit ARIMA$(1,0,0)(0,1,1)_{12}$ directly to the original monthly response series.

(E) Fit ARIMA$(1,1,1)(0,0,0)_{12}$ directly to the original monthly response series.

END OF QUESTIONS

Answer key and explanations

This appendix is for self-marking. To compile questions only, set `\showsolutionsfalse` in the preamble.

1	2	3	4	5	6	7	8	9	10
B	B	D	D	E	E	B	B	A	A
11	12	13	14	15	16	17	18	19	20
E	C	B	D	C	C	E	D	A	C
21	22	23	24	25	26	27	28	29	30
E	A	A	C	C	A	B	E	D	D
31	32	33	34	35	36	37	38	39	40
E	D	D	C	B	A	C	C	A	D
41	42	43	44	45	46	47	48	49	50
E	A	D	B	C	E	B	B	E	A

1. (B) Use a paired t-test on $D_i = \text{after}_i - \text{before}_i$, with $H_0 : \mu_D = 0$. The pairing must be retained; the normality assumption concerns the differences, not each occasion separately.

2. (B) Welch's two-sample t-test permits unequal unknown variances. Its standard error is $\sqrt{s_1^2/n_1 + s_2^2/n_2}$; the pooled method assumes a common variance and paired methods require meaningful pairs.

3. (D) The pooled two-sample t-test uses a common variance estimate and $n_1 + n_2 - 2 = 30$ degrees of freedom. This question specifically asks for that common-variance procedure; it is not a paired design.

4. (D) The interval contains zero, so the corresponding two-sided test does not reject equality. Failure to reject does not demonstrate exact equality.

5. (E) Use the pooled two-proportion z-test of $H_0 : p_1 = p_2$. Pooling estimates the common probability under the null. The response is a pass/fail outcome summarized as independent counts, not a continuous test score.

6. (E) For a normal sample with unknown mean, use $\chi^2 = (n-1)s^2/4$ with $n - 1 = 15$ degrees and an upper-tail alternative. An F variance-ratio test requires two independent samples.

7. (B) Use $F = s_A^2/s_B^2$ with numerator and denominator degrees $(n_A - 1, n_B - 1)$. This tests variability, whereas the pooled two-sample t-test concerns mean differences. Exact variance-ratio inference is sensitive to departures from normality.

8. (B) Use one-way ANOVA for the CRD. Test $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ with $F = MS_{treatment}/MS_{error}$. Rejection establishes that not all means are equal, without identifying the differing pairs.

9. (A) Unadjusted Fisher LSD controls each comparison at its nominal level; it does not generally strongly control the familywise error rate across many comparisons.

10. (A) A small p-value is evidence against the specified null, not a measure of benefit size. Assess the effect, interval, costs, and context.

11. (E) With an intercept, $K - 1 = 3$ indicators suffice. All four indicators sum to the intercept column, making the design matrix rank deficient.

12. (C) $\text{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 comes from regressing X_j on the other predictors, with an intercept. A value of 1 is possible.

13. (B) Adding a predictor cannot decrease training R^2, but adjusted R^2 can fall because it penalizes unnecessary parameters.

14. (D) A funnel suggests heteroscedasticity, namely nonconstant conditional error variance. It does not by itself establish nonlinearity or nonnormality.

15. (C) Durbin–Watson targets first-order error autocorrelation in an appropriate ordered regression setting. Values near 2 suggest little such correlation; formal inference depends on the design and test assumptions. It is not a test of normality or homoscedasticity.

16. (C) The test set has influenced selection, so it is no longer an untouched final evaluation. Use training/validation or cross-validation for selection and a separate final test set.

17. (E) Ridge uses an L_2 penalty and usually shrinks coefficients without setting them exactly to zero. Lasso uses an L_1 penalty and can select variables.

18. (D) The normal equations give $\widehat{\beta} = (X^\top X)^{-1} X^\top y$. Full column rank ensures the inverse exists; dimensions must also conform.

19. (A) The analysis has multiple responses and multiple predictors, hence multivariate multiple regression. Binary logistic regression has one binary response, not a pair of continuous response variables.

20. (C) The mean $E(Y \mid X) = \beta_0 + \beta_1 X + \beta_2 X^2$ is quadratic in speed but linear in the coefficients. Response type and error assumptions support a polynomial normal linear model, rather than a binary, count or ordinal response model.

21. (E) A GLM specifies $g(\mu_i) = x_i^\top \beta$ for $\mu_i = E(Y_i \mid x_i)$. It does not generally model $g(Y_i)$ by ordinary least squares.

22. (A) Binary logistic regression directly models $\log[p/(1-p)] = X\beta$ for the single binary event failure. It does not require multivariate normal predictors within classes; its fitted event probabilities remain between 0 and 1.

23. (A) $p = e^\eta/(1 + e^\eta) = 3/4 = 0.75$. The value 3 is the odds, not the probability.

24. (C) $e^\beta = 1.5$ multiplies the odds by 1.5. The probability change depends on the starting probability.

25. (C) Exponentiate the endpoints: $(e^{-0.2}, e^{0.6}) \approx (0.819, 1.822)$. The interval contains 1, matching a coefficient interval containing 0.

26. (A) The LRT statistic is $120 - 112 = 8$, with 2 added parameters as degrees of freedom, under the usual regularity assumptions.

27. (B) Hosmer–Lemeshow compares grouped observed and expected event counts. A nonsignificant result means insufficient evidence of the grouped discrepancy tested; it does not establish perfect calibration or high classification accuracy.

28. (E) Every observation already classified positive remains positive, and additional observations may become positive. Sensitivity cannot decrease, while specificity cannot increase on that fixed dataset.

29. (D) Naive Bayes factors $P(x_1, \ldots, x_p \mid C)$ into $\prod_j P(x_j \mid C)$. Marginal independence is not required.

30. (D) The estimate is $(0+1)/(5+3) = 1/8$. The denominator includes one added count for each possible predictor level.

31. (E) $P(B \mid D) = 0.4(0.05)/[0.6(0.02) + 0.4(0.05)] = 0.020/0.032 = 0.625$.

32. (D) LDA combines class-specific Gaussian means, a common covariance matrix and class priors via Bayes' theorem. QDA relaxes the common-covariance assumption; Naive Bayes adds a conditional-independence restriction. PCA is not itself a supervised classifier.

33. (D) QDA estimates separate class covariance matrices and generally gives quadratic decision boundaries. Adequate class sample sizes matter because these extra parameters increase estimation variance; greater flexibility does not guarantee better held-out accuracy.

34. (C) An unordered four-category response supports baseline-category multinomial logistic regression with $K - 1 = 3$ comparison equations. Ordinal regression imposes an order; binary collapse would discard the distinctions the researcher wants to model.

35. (B) The coefficient controls $\log(\pi_{bus}/\pi_{car})$. Other category logits also affect normalization, so it need not imply an increase in the absolute bus probability.

36. (A) The proportional-odds ordinal model uses the known ordering and common slopes across cumulative logits. For $\log\{P(Y \leq k)/P(Y > k)\} = \tau_k - X\beta$, a positive slope shifts probability towards higher categories. Equal category spacing is not required.

37. (C) Use Poisson regression with $\log \mu = X\beta$. It permits zero counts and gives positive conditional means; basic Poisson inference assumes conditional variance equal to the conditional mean, not necessarily equality of the pooled marginal moments.

38. (C) A homogeneous Poisson process with rate 4 per hour has Exponential interarrival times with mean 1/4 hour, or 15 minutes. The Poisson distribution models event counts in a fixed interval, rather than the continuous waiting time.

39. (A) A Gamma GLM permits $\text{Var}(Y \mid X) = \phi\mu^2$; here $\phi = 0.4$. The log link preserves positive means. Exponential waiting-time regression is the Gamma shape-one case and imposes $\text{Var}(Y \mid X) = \mu^2$, which does not match 0.4.

40. (D) The fitted conditional mean ratio is $e^{\log(0.8)} = 0.8$, corresponding to a 20% lower expected count, holding other predictors constant.

41. (E) Bayesian logistic regression forms $p(\beta \mid D) \propto L(\beta)p(\beta)$. The priors here describe uncertainty about coefficients; they should not be confused with the class prior probabilities used by a Naive Bayes classifier.

42. (A) This is a posterior probability about the coefficient conditional on the data, likelihood, and prior. It is not a frequentist p-value or a probability increment.

43. (D) Treatments are assigned independently to classrooms, so classrooms are experimental units. Treating all 120 students as independent treatment replications would ignore clustering.

44. (B) Use a randomized complete block design: every soil block contains every fertilizer, randomly assigned to separate plots within that block. Blocking accounts for known nuisance variation; it does not replace randomization.

45. (C) Standardized PCA is unsupervised and constructs orthogonal loading directions with uncorrelated scores, ranked by explained variance. Standardization gives each input variance 1 so different measurement units do not determine the initial scale.

46. (E) Total variance is $5 + 2 + 1 = 8$. The first two explain $(5 + 2)/8 = 0.875 = 87.5\%$.

47. (B) PCA produces uncorrelated component scores and orthogonal loading directions. Uncorrelatedness does not generally imply statistical independence.

48. (B) Multiplicative decomposition $y_t = T_t S_t R_t$ suits positive series whose seasonal amplitude is proportional to the level. An alternative is additive decomposition after a suitable log transformation. Visible calendar seasonality is incompatible with an unadjusted constant-mean stationary model.

49. (E) Expand $(1 - B)(1 - B^{12})y_t$ to obtain $y_t - y_{t-1} - y_{t-12} + y_{t-13}$.

50. (A) The ARMA$(1, 1)$ describes $(1 - B^{12})y_t$, so the original model has $(p, d, q) = (1, 0, 1)$ and $(P, D, Q) = (0, 1, 0)$ at period 12. Fitting that specification to an already seasonally differenced series would apply a second seasonal difference.